SME FINANCE COPILOT: AI AS A FINANCIAL PARTICIPANT

FinTechathon 2026 — Topic E Executive Pitch Deck

- **Local-First Autonomous Financial Operating System** designed for emerging market micro & small enterprises.
- **Zero-Float Deterministic Accounting:** All transactions tracked in 64-bit integer minor currency units.
- **Credit Underwriting & DSCR Risk Engine:** Instant loan packaging for Tier-1 commercial partner banks.
- **11-Domain Grounded Financial AI Copilot:** Real-time loss turnaround diagnosis and margin guardrails.
- **100% Offline Edge Capable:** Tested and verified with 224/224 passing automated regression tests.

THE CORE SME CRISIS & PASSIVE ACCOUNTING FAILURE

Why Small Businesses Need an Active Financial Copilot

- Over 400M global small enterprises operate without professional financial accountants or CFOs.
- More than 80% struggle with cash flow collapse, predatory interest rates, and loan rejections.
- Traditional accounting software is purely passive: records historical entries but provides zero proactive advice.
- Generic LLM Chatbots hallucinate figures and cite irrelevant legal contracts when asked operational questions.
- SME Finance Copilot acts as an autonomous participant: defending cash runways, identifying overhead waste, and unlocking credit.

ARCHITECTURAL PILLARS & NUMERICAL INTEGRITY

Engineering Trust, Precision & Edge Resilience

- **Zero-Float Precision:** Integer minor units eliminate floating-point rounding errors across all ledgers.
- **Local-First Edge Architecture:** Self-contained SQLite WAL engine operating with zero mandatory cloud lock-in.
- **Multi-Tenant Relational Isolation:** Cryptographic HMAC session tokens and strict database query bounds (HTTP 403).
- **Consequential Action Safeguard:** Two-step cryptographic confirmation barrier prevents unauthorized AI modifications.
- **Tamper-Evident Audit Trails:** Cryptographic hash-chained immutable activity logging.

REAL-WORLD TURNAROUND: THE LOSS RECOVERY ENGINE

Diagnosing & Modeling Turnaround from Real Ledger Records

- **Test Failure Mode Repaired:** Merchant asks *"I am in loss how can I recover?"* (Shop #28 Bismillah Store).
- **Grounded Reconciled Actuals:** $1.52M Revenue vs $4.86M Expenses = -$3.34M Net Operating Loss.
- **Root Cause Analysis:** Audits categories on 100% denominator, detects $2.95M facility costs & 82 duplicate entries ($603k).
- **Working Capital Distinction:** Identifies uncollected receivables ($89.3k) and inventory at cost ($2.45M) without treating them as immediate cash.
- **Quantified Scenarios:** 5% expense cut = **$243,408**; 10% sales lift = **$152,540** top-line revenue; combined gross turnaround = **$395,949** (net impact: **$266,061**).

INSTITUTIONAL BRIDGE: INSTANT CREDIT UNDERWRITING

Unlocking Commercial Bank Lending for Unstructured SMEs

- **Real-Time Loan Calculation:** Monthly EMI, amortization schedules, and total interest.
- **Banking DSCR Benchmark:** Automated Debt Service Coverage Ratio evaluation against the 1.25x bank threshold.
- **Partner Bank Matching:** Real-time suitability ranking across 5 commercial banks (BRAC Bank, City Bank).
- **Official Underwriting Packet PDF:** Generates verified credit dossiers with watermarked cryptographic hashes.
- **Smart Document OCR:** Staged human-in-the-loop invoice extraction before general ledger posting.

VERIFICATION, COMPLIANCE & PRODUCTION READINESS

Master Competition Delivery Verification

- **224 / 224 Automated Tests Passed** across 21 test suites in 20.77 seconds (100% pass rate).
- **Formal 50-Test Catalogue (TC-01..TC-50)** covering security, arithmetic bounds, and RAG retrieval.
- **Native Multilingual Intelligence:** Fluent synthesis in English, Bengali (■■■■■), and Chinese (■■■■).
- **Enterprise Currency Preservation:** Configured business currency (USD) strictly preserved across language switches.
- **Disaster Recovery:** Verified SQLite WAL backups (23.46 MB, 163,054 records) with SHA-256 manifests.